

TITLE: IMPLEMENTATION OF BASIC TWO DIMENSIONAL TRANSFORMATIONS

OBJECTIVE

- To understand and implement various 2D geometric transformations and observe how the coordinates of a 2D object change after applying transformation operations.

THEORY:

Two dimensional transformations are used in computer graphics to modify the position, size, and orientation of objects. These transformations are applied on points, lines, and shapes.

A 2D object is represented using coordinate points (x, y). Transformations change these coordinates using mathematical operations.

In this experiment, transformations are applied on a line segment.

The basic 2D transformations are:

1. Translation
2. Scaling
3. Rotation
4. Reflection
5. Shearing

1. Translation:

Translation moves an object from one position to another along a straight line.

If (x, y) is the original point and (tx, ty) is the translation distance, then the new point is:

- $x' = x + tx$
- $y' = y + ty$

Translation does not change the shape or size of the object.

2. Scaling:

Scaling changes the size of an object.

If (sx, sy) are the scaling factors, then the new point is:

- $x' = x * sx$
- $y' = y * sy$

If scaling factor $> 1 \rightarrow$ object enlarges

If scaling factor $< 1 \rightarrow$ object shrinks

3. Rotation

Rotation turns the object about the origin or a fixed point.

If θ is the angle of rotation, then:

- $x' = x \cos\theta - y \sin\theta$
- $y' = x \sin\theta + y \cos\theta$

Rotation can be clockwise or anticlockwise.

4. Reflection

Reflection produces a mirror image of the object.

Reflection about X-axis:

- $x' = x$
- $y' = -y$

Reflection about Y-axis:

- $x' = -x$
- $y' = y$

Reflection about origin:

- $x' = -x$
- $y' = -y$

5. Shearing

Shearing distorts the shape of the object.

Shear in X-direction:

- $x' = x + shx * y$
- $y' = y$

Shear in Y-direction:

- $x' = x$
- $y' = y + shy * x$

ALGORITHM

- Translation:

Step 1: Start

Step 2: Input the coordinates of the object

(x1, y1) and (x2, y2)

Step 3: Input translation distances

tx and ty

Step 4: Calculate new coordinates

$$x1' = x1 + tx$$

$$y1' = y1 + ty$$

$$x2' = x2 + tx$$

$$y2' = y2 + ty$$

Step 5: Display the translated coordinates
(x1', y1') and (x2', y2')

Step 6: Stop

- Scaling:

Step 1: Start

Step 2: Input the coordinates of the object
(x1, y1) and (x2, y2)

Step 3: Input scaling factors
sx and sy

Step 4: Calculate new coordinates
 $x1' = x1 * sx$
 $y1' = y1 * sy$
 $x2' = x2 * sx$
 $y2' = y2 * sy$

Step 5: Display the scaled coordinates
(x1', y1') and (x2', y2')

Step 6: Stop

- Rotation:

Step 1: Start

Step 2: Input the coordinates of the object
(x1, y1) and (x2, y2)

Step 3: Input angle of rotation θ

Step 4: Convert angle into radians
 $rad = \theta \times 3.14 / 180$

Step 5: Calculate new coordinates
 $x1' = x1 \cos(rad) - y1 \sin(rad)$
 $y1' = x1 \sin(rad) + y1 \cos(rad)$
 $x2' = x2 \cos(rad) - y2 \sin(rad)$
 $y2' = x2 \sin(rad) + y2 \cos(rad)$

Step 6: Display the rotated coordinates

Step 7: Stop

- Reflection

Step 1: Start

Step 2: Input the coordinates of the object
(x1, y1) and (x2, y2)

Step 3: Choose reflection type

1. X-axis
2. Y-axis
3. Origin

Step 4: Apply reflection

If reflection about X-axis

$$x' = x$$
$$y' = -y$$

If reflection about Y-axis

$$x' = -x$$

$$y' = y$$

If reflection about Origin

$$x' = -x$$

$$y' = -y$$

Step 5: Display reflected coordinates

Step 6: Stop

- Shearing

Step 1: Start

Step 2: Input the coordinates of the object

(x1, y1) and (x2, y2)

Step 3: Choose shearing type

1. X-direction

2. Y-direction

Step 4: Input shearing factor

shx or shy

Step 5: Apply shearing

If X-direction shear

$$x' = x + shx * y$$

$$y' = y$$

If Y-direction shear

$$x' = x$$

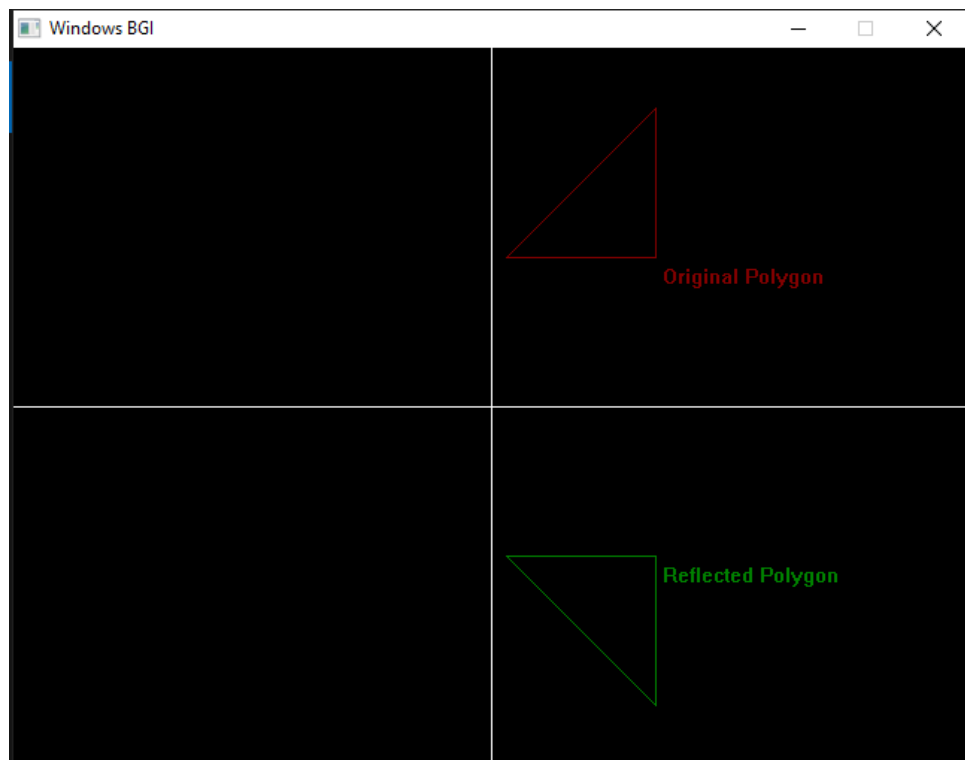
$$y' = y + shy * x$$

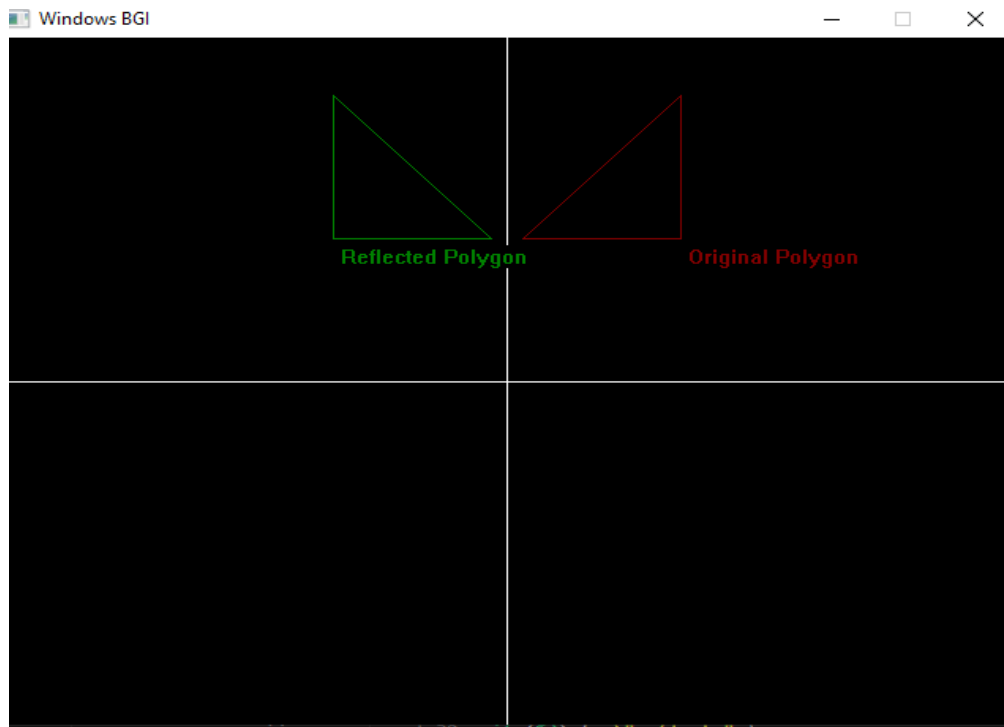
Step 6: Display sheared coordinates

Step 7: Stop

OUTPUT:







CONCLUSION

The experiment was successfully completed. Different 2D transformations were applied on a line segment and the new coordinates were calculated. These transformations are useful in computer graphics for moving, resizing, rotating, and modifying objects.